

Department of Electronics and Electrical Engineering
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EE101 - Basic Electronics

Pre-Tutorial Assignment 3: Solutions

1. Figure 1 shows an electric circuit with independent voltage and current sources. The numerical values of the circuit elements are as follows: $V_1 = 10V$, $V_2 = 5V$, $I_1 = 1A$, $R_1 = 1\Omega$, $R_2 = 2\Omega$, and $R_3 = 3\Omega$. Using mesh analysis, evaluate the potential difference V_{I1} across the current source I_1 .

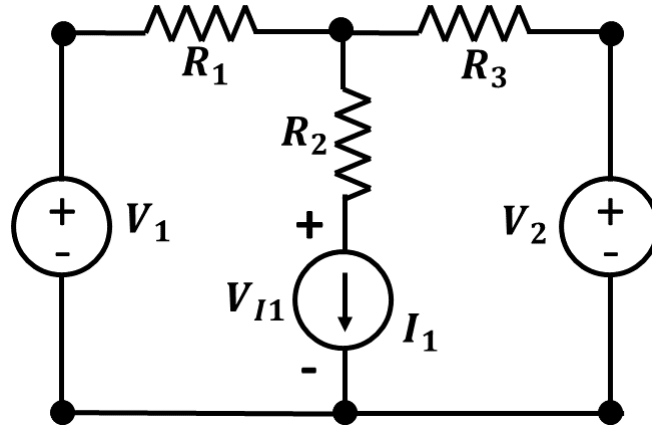


Figure 1: Electric circuit with independent voltage and current sources.

Solution

Figure 1 marks the two meshes M_1 and M_2 . Let the mesh currents in the meshes M_1 and M_2 be i_1 and i_2 respectively.

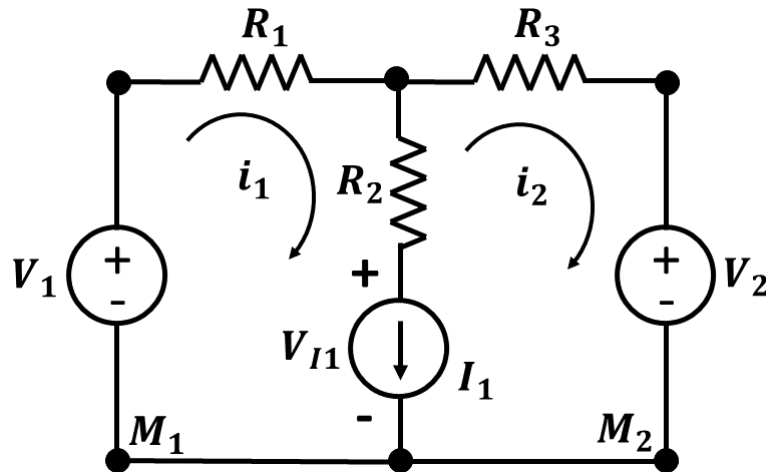


Figure 2: Electric circuit with independent voltage and current source. The meshes M_1 and M_2 and the respective mesh currents i_1 and i_2 are shown in the figure.

The current source I_1 is shared between M_1 and M_2 . Thus, the mesh currents can be related as follows.

$$i_1 - i_2 = I_1 \quad (1)$$

$$i_1 = i_2 + I_1 \quad (2)$$

Application of KVL at mesh $\mathbf{M}_1$ provides us with the following equation.

$$-V_1 + R_1 i_1 + R_2 (i_1 - i_2) + V_{I1} = 0 \quad (3)$$

The following equation is obtained by rearranging terms of the above equation and using equations .

$$R_1 i_2 + V_{I1} = V_1 - (R_1 + R_2) I_1 \quad (4)$$

We obtain the following equation by applying KVL at mesh $\mathbf{M}_2$.

$$-V_{I1} - R_2 (i_1 - i_2) + R_3 i_2 + V_2 = 0 \quad (5)$$

Rearranging terms and substituting $i_1 - i_2 = I_1$ in the above equation provides us with the following equation.

$$R_3 i_2 - V_{I1} = -V_2 + R_2 I_1 \quad (6)$$

Solving equations 4 and 6, we obtain the following.

$$i_2 = \frac{V_1 - V_2 - R_1 I_1}{R_1 + R_3} \quad (7)$$

$$V_{I1} = R_3 i_2 + V_2 - R_2 I_1 \quad (8)$$

Putting numerical values, we obtain

$$i_2 = \frac{V_1 - V_2 - R_1 I_1}{R_1 + R_3} = \left(\frac{10 - 5 - 1 \times 1}{1 + 3} \right) A = 1A \quad (9)$$

$$V_{I1} = R_3 i_2 + V_2 - R_2 I_1 = (3 \times 1 + 5 - 2 \times 1)V = 6V \quad (10)$$

2. Consider a four bit number $(ABCD)_2$ which has a value between 0 and 15, both inclusive. A digital circuit needs to be built which would recognize if the input is a multiple of $(11)_2$. Define the function $F(A, B, C, D)$ so that $F = 1$ if and only if the number $(ABCD)_2$ is divisible by $(11)_2$.
- (a) Express the function $F(A, B, C, D)$ in the Standard SOP form.
- (b) Express the function $F(A, B, C, D)$ in the Standard POS form.

Solution

The truth table for the function $F(A, B, C, D)$ is as shown below:

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

- (a) Standard SOP: $F(A, B, C, D) = \overline{A}.\overline{B}.\overline{C}.\overline{D} + \overline{A}.\overline{B}.C.D + \overline{A}.B.C.\overline{D} + A.\overline{B}.\overline{C}.D + A.B.\overline{C}.\overline{D} + A.B.C.D$
- (b) Standard POS: $F(A, B, C, D) = (A+B+C+\overline{D})(A+B+\overline{C}+D)(A+\overline{B}+C+D)(A+\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+\overline{D})(\overline{A}+B+C+D)(\overline{A}+B+\overline{C}+D)(\overline{A}+B+\overline{C}+\overline{D})(\overline{A}+\overline{B}+C+\overline{D})(\overline{A}+\overline{B}+\overline{C}+D)$